Portfolio of Evidence (PoE) - Week 1 Decision Log & Runbooks

Student Name: Zuki Zihlangu

Cloud Provider: Amazon Web Services (AWS)

Primary Region: Africa (Cape Town) af-south-1

Project: WIL-2026 Cloud Infrastructure Setup

1. Architectural & Security Decision Log

Root Account Hygiene & Identity Protection

- **Decision:** Enforce Multi-Factor Authentication (MFA) on the root user account immediately after account creation.
- **Justification:** The AWS root account possesses unrestricted access to all resources and billing data. Utilizing hardware or virtual MFA mitigates credential theft risks, preventing unauthorized access even in the event of password exposure. Root operations are restricted strictly to initial configuration and billing setup.

Cost Governance & Budget Alerting

- **Decision:** Provision an AWS Billing Budget capped at $5.00 USD with automated email notifications.
- **Justification:** Financial governance is critical in cloud operations. Implementing zero-spend and low-threshold alarms ensures immediate awareness of rogue resource allocation or unexpected charge spikes, maintaining strict adherence to free-tier and educational budget bounds.

Regional Infrastructure Selection (af-south-1)

- **Decision:** Standardize all resource deployments in the Cape Town (af-south-1) region.
- **Justification:** Localizing infrastructure within South Africa satisfies data sovereignty compliance standards, minimizes network latency for regional client traffic, and ensures accurate performance testing for local cloud architecture workloads.

Object Storage Immutability & Encryption Baseline

- **Decision:** Enforce S3 Bucket Versioning, Server-Side Encryption (SSE-S3), and explicit Block Public Access.
- **Justification:**
- **Versioning:** Secures object state history to protect critical data against accidental deletion, malicious overwrites, or ransomware incidents.
- **Encryption:** Automatically encrypts data at rest using AES-256 managed keys (SSE-S3) at no additional administrative cost.
- **Block Public Access:** Enforces an absolute perimeter defense, eliminating accidental public bucket exposure at the API and policy level.

Perimeter Network Defense (Least Privilege)

- **Decision:** Restrict Security Group (wil-zuki-edge-sg) inbound traffic exclusively to standard web ports (HTTP: 80, HTTPS: 443).
- **Justification:** Following the principle of least privilege, public access is permitted strictly for web traffic. Sensitive administrative ports (SSH: 22, RDP: 3389) are deliberately excluded from public exposure (0.0.0.0/0) to minimize attack vectors and prevent brute-force remote shell access attempts.

2. Technical Comparison Matrix: SG vs. NACL

Feature/ Property	Security Group (SG)	Network Access Control List (NACL)
Operating Layer	Elastic Network interface (ENI) / instance Level	Subnet Boundary Level
Traffic State	Stateful (inbound return traffic automatically allowed)	Stateless (Inbound/Outbound traffic must be explicitly defined
Rule Evaluation	Evaluates all rules before granting/denying access	Evaluates rules in strict numerical order (lowest number first)
Rule Action	Supports Allows rules only	Supports both Allow and Deny rules

3. Operational Runbook: S3 Object Version Restoration

Objective

Restore a previous version of a file stored in an S3 bucket (wil-zuki-datavault-2026) following an accidental overwrite or deletion event.

Step-by-Step Recovery Procedure

Navigate to the **Amazon S3 Console** and select the target bucket (wil-zuki-datavault-2026).

Open the **Objects** tab.

Toggle the **Show versions** switch to **ON** to expand the complete version history of all objects.

Locate the object requiring restoration and identify the top row entry (marked as the current/latest version).

Select the checkbox for the current version and click **Delete**.

Confirm deletion by typing permanently delete.

Verification: Upon deletion of the top version, AWS S3 automatically promotes the immediate prior version to become the active, live object.

4. Portfolio of Evidence File Index

- **01-Data-Vault/**

- Day2-s3-settings.png - Verification of S3 Block Public Access, Versioning, Encryption, and Project Tags.
- Day3-restore-success.png - S3 Versioning rollback and restore verification.
- **02-Secure-Edge/**
- Day4-sg-rules.png - Least-privilege inbound security group configuration (wil-zuki-edge-sg).
- **03-Ops-Automation/**
- Mfa-status.png - Account root MFA configuration proof.
- billing-budget-alarm.png - AWS Billing Budget and alarm trigger setup ($5 limit).
- Week1-aws-learning-platform.png - AWS Skill Builder / Educate profile completion.
- **Decisions/**
- decision-log.md - Complete architectural justifications, SG/NACL matrix, and S3 restore runbook.
- **LinkedIn/**
- Week1-post.png - Published LinkedIn post summarizing Week 1 accomplishments and tag requirements.

Evidence Links and resource index

1. **S3 Bucket URL**
   wil-zuki-datavault-2026 - S3 bucket | S3 | af-south-1

2. **Security Group URL**
   Security groups | EC2 | af-south-1

3. **Billing Budget URL**
   Budget details | AWS Budgets | Global